Math 4141: Geometry and Differential Calculus

Dr. Md Tusher Mollah

Assistant Professor, Department of Natural Sciences
Islamic University of Technology, Bangladesh
PhD and Postdoc from the Department of Civil and Mechanical Engineering

BSMRSTU, Bangladesh
- B.Sc. (2013-2016)
- M.Sc. (2017-2018)

CUST, Bangladesh
- Adjunct Lecturer (2018-2019)

EUB, Bangladesh
- Lecturer (2019)

LU, UK
- Visiting researcher (2022)

DTU, Denmark
- PhD (2019-2023)
- Research Assistant (2023)
- Postdoc (2023-2025)

IUT, Bangladesh
- Assistant Professor

Math 4141: Geometry and Differential Calculus **Credit: 4.00**

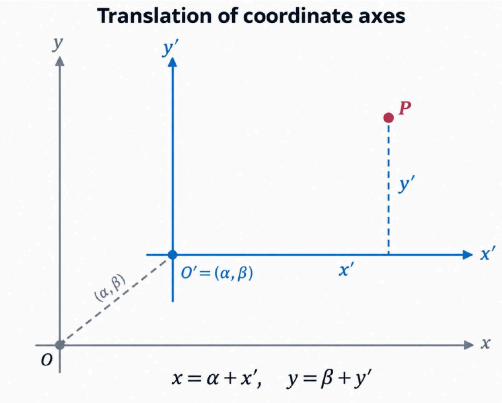
Course Contents:

2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.

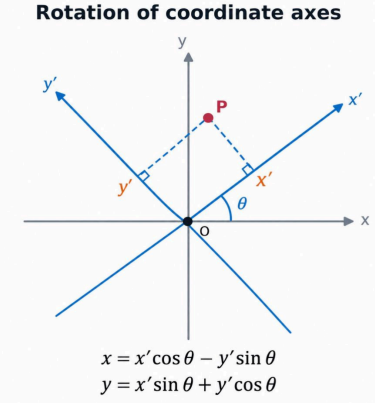
3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing.

Change of Axes:



$x = \alpha + x', \quad y = \beta + y'$



$x = x'\cos\theta - y'\sin\theta$
$y = x'\sin\theta + y'\cos\theta$

Change of Axes: (Exercise)

Q. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin transferred to the point $(2, -1)$.

Solution: Given $2x^2 + 3y^2 - 8x + 6y - 7 = 0$……(1)

If the origin transferred to the point $(2, -1)$

Put $x = \alpha + x' = 2 + x'$

and $y = \beta + y' = -1 + y'$

From (1) we get $2x'^2 + 3y'^2 = 18$

Now removing suffixes $2x^2 + 3y^2 = 18$

Q. Determine the equation of the parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ after rotating axes through 45^0.

Solution: Given $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$……..(1)

If the axes have been rotating 45^0

$$x = x'\cos\theta - y'\sin\theta = x'\cos 45^0 - y'\sin 45^0 = \frac{x' - y'}{\sqrt{2}}$$

and $y = x'\sin\theta + y'\cos\theta = x'\sin 45^0 + y'\cos 45^0 = \frac{x'+y'}{\sqrt{2}}$

From (1) we get $2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0$

Now ignoring suffixes $2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$.

Change of Axes: (Exercise)

Q. By transforming to parallel axes through a properly chosen point (h, k). Prove that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ can be reduced to one containing only the terms of the 2nd degree.

Solution: Given $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$……(1)

If transforming axes through (h, k)

Put $x = \alpha + x' = h + x'$

and $y = \beta + y' = k + y'$

Form equation (1) we get,

$12x'^2 - 10x'y' + 2y'^2 + (24h - 10k + 11)x' + (-10h + 4k - 5)y' + 12h^2 - 10hk + 2k^2 + 11h - 5k + 2 = 0$……..(2)

Now equate the coefficient of x' and y' we get

$24h - 10k + 11 = 0$……(3)

$-10h + 4k - 5 = 0$……..(4)

After solving $h = -\frac{3}{2}$ and $k = -\frac{5}{2}$

From equation (2) we get,

$12x'^2 - 10x'y' + 2y'^2 + 0.x' + 0.y' + 12(-\frac{3}{2})^2 - 10\left(-\frac{3}{2}\right)(-\frac{5}{2}) + 2\left(-\frac{5}{2}\right)^2 + 11(-\frac{3}{2}) - 5(-\frac{5}{2}) + 2 = 0$

$\therefore 12x'^2 - 10x'y' + 2y'^2 = 0$